

AYMANE AIT DADS

AI Machine Learning Intern | Data Science · Machine Learning · Predictive Analytics

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PROFESSIONAL SUMMARY

Data Science engineering student who reduced manual network analysis time by ~60% at Orange Maroc by building automated ML pipelines on 100K+ records, delivering actionable maintenance recommendations directly to the network optimization team. Ranked 1st in class on an AI-image detection benchmark (0.791 AUC) and achieved 99.8% accuracy on aerial species classification, demonstrating strong command of model evaluation, ensemble methods, and data-driven decision-making. Brings expertise in Python, SQL, Scikit-learn, Pandas, Power BI, and feature engineering aligned with SAP's need for ML talent capable of translating model outputs into business insights across large-scale enterprise data. Aims to apply machine learning and data analysis skills within SAP's iXp program to help build intelligent, industry-grade AI solutions that support smarter enterprise operations.

CORE COMPETENCIES

Machine Learning | Data Analysis | Predictive Analytics | Feature Engineering | Model Evaluation | Data Visualization | Natural Language Processing | Python Programming

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built a **Random Forest + K-Means machine learning pipeline** on 100K+ fiber-optic network records - including upload speed, latency, and jitter - to classify network quality across thousands of nodes.
- Developed a clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for infrastructure decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering and model evaluation workflows, accelerating the reporting cycle for network quality assessments.
- Delivered **stakeholder presentations** using Power BI to translate model outputs into maintenance recommendations, bridging technical findings and business decision-making for non-technical audiences.

PROJECTS

Aerial Cactus Detection - Endangered Species Classification

EURECOM · 2025 · 99.8% Accuracy

- Benchmarked CNN, DenseNet121, and hybrid CNN-Transformer architectures on **17,500+ drone images**, achieving 99.8% accuracy, F1 = 0.999, and ROC AUC = 1.0 for endangered species classification.
- Applied systematic **model evaluation** across architectures, selecting the optimal backbone through ablation and validating generalization with held-out test sets, producing results fit for conservation reporting.

PyTorch · DenseNet121 · CNN · Scikit-learn · Kaggle

NVIDIA Nemotron Model Reasoning Challenge

Kaggle · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter** Mamba-Transformer with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context).
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers for verified chain-of-thought data.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, Computer Vision, NLP, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

Data Science & Analytics: Python | SQL | Pandas | NumPy | Power BI | Scikit-learn | Feature Engineering

Machine Learning & Deep Learning: PyTorch | TensorFlow | Hugging Face Transformers | LoRA / PEFT | Ensemble Methods | NLP

Tools & Infrastructure: Git | Linux | Docker (basic) | Kaggle GPU | FP16/BF16 Mixed Precision | REST APIs